Stacy Rasgon: “Demand Is Off The Charts” in Semiconductors… And Set To Double Again Soon

Podcast: Monetary matters

AI Executive Summary

1. Core Thesis: The semiconductor sector is currently in a structural bull phase driven by genuine, unprecedented demand from the AI revolution rather than a typical cyclical peak. While earnings multiples have compressed and channel inventories remain high, these are not bear signals; instead, they reflect supply chain bottlenecks (clean rooms, power) that will support growth for 2027-2028. Fundamentals remain strong due to hyperscalers' inability to stockpile AI capacity immediately, with demand visibility outpacing production capabilities across memory, logic, and capital equipment sectors.

2. Key Arguments & Data:

3. **Earnings vs. Valuations:** Semiconductor earnings are up approximately 100% year-to-date while stocks remain below June highs; forward P/E multiples have compressed due to growth expectations remaining higher than current prices.
4. **Demand Reality:** Major players (Nvidia, Broadcom) guide high growth (70-100%), yet investors worry about peak timing. The consensus among industry CEOs is that demand is strong and supply constraints are limiting output, not limiting adoption.
5. **Memory Cycle Squeeze:** DRAM/NAND supply remains tight due to AI hardware requirements. HBM (High Bandwidth Memory) consumes 3-4x more wafer capacity than standard DRAM per gigabyte. AI infrastructure racks use significant memory (up to 80% wafer area inside a rack).
6. **Capital Equipment (Semi-Cap):** Wafer Fabrication Equipment (WFE) spend is forecast at $150B in 2026, rising to $200B+ by 2027 and over $250B by 2028.
7. **Clean Room Constraint:** A major supply bottleneck exists; manufacturers cannot ship wafers until they build dedicated clean rooms. WFE growth is currently capped by facility availability, not equipment manufacturing capacity.
8. **Inventory Anomalies:** Channel inventory remains high (historically a red flag), but it has been elevated for 4-5 years and does not signal a

peak in this cycle due to AI-driven demand outstripping stockpiling capability.

9. **Market Cap Dynamics:** While Nvidia and Broadcom have seen lags, the market treats them as stable cash flows for shorting other sectors. They maintain dominance via CUDA ecosystem and balance sheet strength, limiting risk of immediate loss of share.
10. **Intel Turnaround:** Intel has shown signs of recovery through server product sales (200 bps gross margin upside) and cost cutting under Lee Gellinger, though it is viewed as a "slog" rather than a full rebound.

11. **Memory Tech Distinction:** DRAM is volatile; NAND is non-volatile but denser per bit. AI usage increases both demand types, particularly HBM for compute logic and NVMe/NAND for storage/weights (KV Cache).

12. Actionable Takeaways:

13. **Monitor "Real" vs. Phantom Demand:** Investors should prioritize capital allocation to companies building infrastructure (AI data centers) over speculative bets on sectors not supporting the AI compute base.
14. **Diversify via Tangibles:** To mitigate potential sector rotation or inflationary risks, investors are encouraged to consider exposure to energy, materials, and precious metals through a "Return to Tangibles" strategy (e.g., LENS ETF), as this provides diversification against bond/stock declines during cycles.
15. **Watch Clean Room Supply:** The most immediate macro constraint for the AI infrastructure cycle is clean room availability. Expect semiconductor capex outflow to be limited by facility build-out rather than pure chip fabrication capacity until 2027-28.
16. **Adjust Inventory Metrics:** Historical cyclical inventory signals may not apply to this era of sustained AI demand. Investors should focus on forward revenue estimates from hyperscalers rather than backward channel inventory data when assessing the cycle peak risk.
17. **Long-Term AI Infrastructure Bet:** With capacity expected to run out before the next generation (post-2028), positions in semi-cap

equipment, memory manufacturers, and infrastructure providers are viewed as bullish for the long term (through 2030).

Full Transcript

This episode is brought to you by Sarmaya Partners, who manages the Return to Tangibles ETF, LENS, ticker L-E-N-S, the Sarmaya Thematics ETF, built for a return to tangibles. Learn more at SarmayaETF.com and see the show notes for disclosures. Let's get into it. Got a very special conversation I'm joined by Stacey Raskin, Managing Director and Senior Analyst of US Semiconductors and Semiconductor Capital Equipment at Bernstein Research, obviously with Semiconductors. powering AI. This is an extremely important topic. Stacy is very in demand. Really appreciate you coming on, Stacy. Thank you. Oh, it's good to be here. Thank you. Stacy, you want to start with a little conundrum we have in the market. There's some things I'm having trouble understanding or squaring, which is that the earnings in the semiconductor sector are up about 100% roughly, but the sector's earnings multiple is down. And as you know, semiconductor stocks are still well below their June highs.

What does the market know or maybe fear that the numbers and the estimates aren't showing? I mean, look, semiconductors typically are a cyclical industry and investors tend to think about that as the stocks are moving. And as investors start to worry that we're getting closer to a peak, the beats on the earnings front tend to get rewarded less and less. And so you start to see the multiples come down. And so that is not an atypical behavior. if you're approaching a cycle peak, and I think that's what the stocks in some sense are telling you that we may be approaching. Now, that being said, it sort of begs the question, what if we're not? And I would argue that at least at this point, we're from an earnings standpoint, we probably aren't. And I'd also say there's a little more angst out there simply because you mentioned the stocks are down quite a bit off the June peak, and although we're still up 80% year to date, like if we'd just gotten here monotonically, everybody would feel fine.

That never happens though and clearly if you piled into the space in the middle of June, you're still hurting. And so these are the kinds of things that sort of drive sentiment and everything else. Sentiment I think in general has been a little worse because of that. That being said, when I look at things, to

me at least fundamentals still look like they're getting better, not worse. As you mentioned, earnings are up tremendously. So if the stocks are up, and when I say stocks in this context, I'm talking about the stocks index. We use that as a general proxy. for the space. It's an index made up of semiconductor and semi-cap companies in. It's up roughly 80%, maybe even a little more a year to date. And the earnings are up more than 100%, but multiples are down. And even since June, as the stocks are down off the peak, the forward earnings estimates since June are actually still up and the stocks are down. So the multiples have compressed even more. But as you start to look out, At least commentary from the different companies. There are companies like NVIDIA and Broadcom who have both said they're gonna still, they think they can grow 70 to 100% plus next year. Broadcom suggests that they can even grow 100% again in 28. You listen to some of the memory companies who have suggested that supply and demand will be even tighter in 27 and 28 than they are in 26. And supply demand for memory is very, very tight in 26.

you look at the general commentary from the companies out there that are actually spending the money on this, the hyperscalers and everything, and the one sort of dominating theme we're hearing from everybody is nobody has enough compute. At this point, fundamentals seem better or not worse. I'm not really worried about a cycle peak yet. I will say the same thing that I've said all along since it started, you know, with your chat GPD showed up at the end of 22. And, you know, at some point, presumably we'll hit a peak and you'll roll over and we'll hit an air pocket or something, but it's not yet. It's not this year. It's not that next year Doesn't look like it's 28 29. I don't know right but my guess is we've still got some room and especially after the pullback and we're going to earnings and It feels to me like things still have more room to run off of this base and a minimum of these fundamentals are still getting better so the fundamentals are still looking solid your estimates are Still high and raising and the same is true of the street Yeah broadly going up and up and up. Yeah, yeah Stacy tell us about the fundamentals that you're seeing I know you just you know went on a road show talking to all the big semiconductor companies What are their base case what they're guiding to?

And where if they outperform or underperform those benchmarks, what would that cause of that be? So I guess was kind of the upside and downside to what you're hearing from the semiconductor community. Yeah. You bet. And so you're right. We did just go on a road show. I do this once a year up in Silicon Valley in September. And we think we saw 11 companies over two days. And I mean, look, the universal sort of takeaway from everybody is everybody sounds great, right? Demand is off the charts and visibility is triggered by the. I understand why that worries people. My general belief is that semiconductor company manager needs actual visibility. What's really going on is zero. They don't know. They're the back of the supply chain. What they see are the orders in front of their face. That's all. I never try to fool myself that it's anything different. Right now, however, their order of visibility is very, very strong. And again, it seems to be strengthening, regardless of who you're talking to and which part of the industry that they're in, things seem to be getting...

better. And like I said, if you talk to an old semi guy and there's a lot of gray hairs in this in this space, that sort of perspective makes us nervous. And he was seeing that in the stocks. Like, as I said, this is why nobody wants to believe it. Right. I'd also say that you have to remember the numbers themselves are getting so big. It just gives you pause. Like, I mean, you take Jensen Wine, who's the CEO of NVIDIA. And a couple of years ago, he was talking about By the end of the decade, the industry will be spending 3 or 4 trillion a year on infrastructure for AI. It sounded like the ravings of a madman at the time. It doesn't sound so crazy anymore, right? We're on that track. I mean, the top five hyperscalers will probably spend 1.5 trillion next year. It's not as nuts, like maybe Jensen was right, but I mean, again, you start to stare at these kinds of numbers and just the sheer magnitude, it makes people nervous. But like I said, the ones that are nervous right now are investors.

The companies that are actually out there playing in this space, investing in it, using it, do not, at this point, sound worried. If anything, they sound more positive with every passing day. So we'll see if that continues or not. Again, like I said, I understand why even that makes people nervous. But right now, demand just looks really strong across the board in most of the AI-related spaces. I happen to be listening to a lamb investor day, and this

guy sounded like a very conservative person, but he said, like, literally the best we've ever. seen in terms of the demand signal. So I'm sure you and the other analysts are encountering that sentiment often. Oh, yeah. Yeah. And again, you have to remember, you don't deploy hundreds of billions or even trillions of dollars on a whim, right? It takes a lot of careful planning and planning not just with one company, but planning across multiple companies across the supply chain because of that.

You're getting orders and things that are being put in place farther than this because everybody has to be able to plan toward this common goal. So naturally the worry is if the demand isn't real then everybody falls together like we're all collectively screwed right that would be the worry. At the same time though you could also argue that if the demand is real and the demand itself does prove to be real we can discuss whether or not there is a return on that demand or not but the demand itself at this point does really seem to be real. The fact that there are constraints And it's actually metering how quickly this can get built up. You could argue that actually may be a positive that it stops the industry from going as nuts as they might want to otherwise go and makes them more measured in terms of the pace of the build out and actually may actually help this be stronger for longer. We're probably seeing some of that as well. And sorry, Stacy, what exactly could give pause on the hyperscalers spending so much?

Oh, I was just saying at some point you could argue if they're spending all this money and there's no return. Yes. That's what it would be. But even that question, I would argue that it's still early. But the incremental data points around is there a return or not on these investments, I think is landing pretty much decisively on the side of yes, there is a return. You've got the neoclouds and others who are renting out capacity and they're renting this out at 30 billion a gigawatt or whatever it is, which I mean puts the payback at. you know, a year, a year and a half for the infrastructure. We're seeing companies, you know, some of the foundational model companies who have workloads and they read that stuff out and some of these guys, they're revenues have gone vertical. You had other cases. You had a meta, for example, a week or two ago, announced news, which is an agent for consumers. And I mean, it's, to me, it was one of the first examples of an actual use case.

for agentic AI that a normal person could use, right? That's both usable and useful. You have to remember like, we all in this, we live in a bubble, right? You're aware of everything that's going on is I always look at people like my wife or my family members, they have no idea what any of this stuff is. They don't know the different model. They don't know what an AI agent is or what it could do. And you get things like as muse and other things like that become more into the broader public consciousness and they allow more of the normies to actually get an idea of what AI can do. I mean, we're still very early in terms of adoption. But clearly at the end of the day, they're spending a lot of money. Somebody has to be willing to pay for it. If it turns out nobody's willing to pay for it, that becomes a problem. I don't think that's where we're going. But ultimately, every AI bear case that you can think of, ultimately, I think, you know, boils down to that answering that question. Is there a return or is there not? I think there is. Earlier, Stacey, you said that, you know, as a...

old hand in the semiconductor industry, you don't put much stock in semiconductor CEOs, their ability to forecast demand. They see the actual orders coming in. Or my ability, or anybody's ability. I'm not blaming them. I would say forecasting in general is an unsolved problem in this space. Everybody's bad. It wasn't that long ago that the whole industry was saying, you know, by 2030, the industry will be doing a trillion dollars in revenue. We're going to do a trillion and a half this year, right? So nobody in the industry goes out of forecast. It's not just the CEOs of semiconductor companies. I'm not blaming it. It's not like it's bad. Like it just is. Yes. And it's a phenomenon of overordering in a bull market. Like if I'm not getting my allocation, I want to order maybe twice as much as what I get. And then when I get it, I just cut off entirely. So the peaks and values are very extreme. They can be. So there's a number of things that can drive semiconductor cycles. You discussed one of them. This is called double ordering. So what happens?

Typically when supply is tight and lead times are lead times are, you know, when I order a part, how long does it take to actually receive it? When lead times start to stretch out and customers can't get what they want, they order more. Just as an example, if you're making widgets and you need a hundred semiconductors and you go to your supplier and they say, okay, I'll give you

10 of them now and I'll give you the other 90 in a year. This happened like during COVID in particular, this happened, right? Lead time stretch. So your next step is to go out and order a thousand semiconductors from anywhere you can possibly source them. on the hope that you can cobble together the hundred that you need and then you cancel all the other orders. And to the industry, they can't tell. So that hundred units of actual demand to the industry collectively looks like a thousand units of demand. They don't know, right? There's no way they could know. There's a variety of other things that can happen that cause cycles, but yeah. So this is what happens when supplies, that's one of the big questions right now, supply in AI is tight.

We talked earlier about people worried that we're approaching a peak. That would be one way. Or do we actually need all of this supply that were all these data centers that we're building out? I would argue the answer at least for now is probably yes. So right now, the other thing you see in this sort of a period is you stockpile all this stuff. Nobody is really stockpiling AI hard where everything that's actually getting built and deployed is being used immediately. So that's a good sign. It's not like during COVID, When supply got really tight and the supply chains got all screwed up because of all the shutdowns and people absolutely were stuck like rich and you could measure it. You can see the inventories. I don't think we're seeing that with AI. The capacity that's actually getting built is getting used to tell me about you said there's not as much stockpiling. Could you give me some color on that and.

You know you you hear you encounter these narratives of well Microsoft or alphabet bought all these chips and they're just lying on the floor because the data center isn't built yet either because there's delays or because now there's a big backlash against data centers as you know What's your color or numbers on why you think that that narrative is a little I do not think that there are like huge warehouses all over the place filled with GPUs that are just I do not think that that is true You talked about backlashes. There are backlashes against data center build-ups in some places. I think in terms of the total amount of capacity that's going to be built, it's not that big. My worry for that, by the way, we could talk about it, is it does make a really good political talking point, backlashing. And it's not just, it's bipartisan, by

the way. It's not just the Democrats or the Republicans. It's both, we are going in the midterms. So that is a, you know, I guess a concern, although my guess is progress will not be stopped. There's enough money and power behind this that I think broadly that they will give it. And we got a lot of land and other, there's a lot of places we can build.

The other thing you'll remember is we don't just build data centers in the US. So if we're not going to build them and we need the capacity, we will build them somewhere else. And then finally, you've got folks like SpaceX and Musk who wants to put all the stuff in orbit. That's why you laugh. I mean, this is one of the reasons, though. There's no nimbyism in space. If you can get it out there and it's viable, you can see from my background, by the way, I'm more than a passing interest in outer space. We can talk about Orville data centers, but that's one reason he wants to put that stuff up there as well. But that is a concern. Power is a big constraint. If you were to ask me, you know, if Jensen says we're going to spend three or four trillion dollars a year, and maybe we will, and he said there was absolutely demand for it, what would stop us from getting there? Power is a huge constraint. Sure. You know, and if you're adding tens or even hundreds of gigawatts a year, this is something like the electrical grids can't really support this.

Yeah, we're doing a lot of what's called behind the meter power power that's actually on site generation. But those are those are things again that have to get worked out through the supply chain. So you always have that sort of stuff. And there's always going to be some amount of capacity, maybe that shipped, but not online yet because we're waiting for the power. But I don't think it's a broad based kind of thing. It's more timing issues on some of the bills. And I don't think it's slowing anything down. And if you look at Nvidia Broadcom who've given pretty strong forecast, Nvidia said that they think they can grow their revenue 70% next year at least. And they've got enough demand to grow 100. Broadcom says they're gonna grow revenues 100% next year and 100% the year after. And they've got demand for more. Like why do you even video say 70% instead of 100%? The 70% is where they believe there will be enough land power and shell that actually gets built to support what they should, that's 70%. They will ship. They think that there's enough of those buildouts that will happen of the infrastructure.

to support that. They're not guiding higher than that because they're not sure if they can get it. If they can get land power and shell that's built out above that, they will take their numbers up. But they are committing to those numbers because they believe that there is enough. They've got line of sight in terms of the projects to have confidence that that stuff will get built. Clearly, if it doesn't get built, no problems. Sure. But they've already tried to haircut and commit to what they believe is viable to get built over the next year or so. When they when they raise their numbers, then you raise your. estimates and then other analysts raise their estimates. And suddenly that's what's in the forward PE, which the analysts, everyone on TV, let me, the podcast we all talk about. Yeah. And by the way, that's not even the number that matters for you. What really matters is not my numbers or my competitors' numbers, it's the investors, the buy side. So if NVIDIA says they're going to grow 70%, I guarantee you the buy side's higher than that. Right. And so the stock, which right now, I mean, if just to pick on NVIDIA, if they grow, Revenue 70% they'll probably do 15 or 16 earnings next year and the stock as of right now if I look at my chart here is Sitting at 240 So on you know 1450 it's like teens, right? If they grow 85% they'll do 17 or 18 bucks if they actually double the revenues They'll probably do 20 in earnings and the stock would be You know, whatever that is drug 11 or 12 times broad broad come suggested by 28 they will do over 30 bucks in earnings, which if you think that they can double the revenues next year and double them again, yeah, you'll be over 30 bucks in earnings. Broadcom stock right now is at 363, so if they do, same thing, it'll be 11 or 12 times earnings on that number if they can do it. So the question then becomes again, back to our earlier discussion, does that represent a peak or not? If it's not a peak, the stocks are absurdly cheap, right? If it is a peak, then maybe they're not, and the market right now is telling you that at least for those kinds of names, they're worried about peaks. You see this in memory as well. A lot of the memory stocks are trading at single digit PEs, right? Because ASPs are up a gazillion percent over the last year and capacity is coming online for memory. So people worry that we're at a peak. Now, Micron just said, yes, capacity is coming online. And by the way, supply is going to be even tighter in 27 or 28 than it is now. So that becomes the question of demand, is that supply comes online, is the demand there to meet it or not?

That gets you back to our earlier point on double ordering. Is it real demand or is it phantom demand? The industry believes and the companies believe that the demand is real. They are now building to that. And if they're right, we won't be at a peak. If they're wrong, then we will and we'll have problems, not just in semiconductors. But let's be honest, I think AI has gotten so big, it's dragging everything along with it in and out of semis. I think the broader like global economy outside of AI is not so great. Hey everyone, hope you're finding value in my questions. I've got a quick question for you. How much of your portfolio actually depends on the physical world? If you own the S&P 500, more than a third is 10. Energy, materials and precious metals about 5%.

Sarmaya Partners believes the next cycle belongs to tangibles. The energy and metal sectors have been under-invested and under-loved for a decade. They call it Return to Tangibles, and LENS, ticker LENS, the Sarmayar ETF, is an actively managed ETF that they manage to express this view. And active is the point. The team follows the theme across sectors, market caps, and borders as the cycle evolves, adding where they see opportunity and trimming themes that look overextended. Miners, energy, precious metals, critical metals, all in one actively managed ticker, LENS. When inflation runs, stocks and bonds can fall together. LENS adds real asset exposure with the potential to diversify both the stock side and the bond side against inflation and geopolitical risk. Learn more about the LENS ETF and their full data and comparison, including performance at SarmayaETF.com.

Investing involves risk, including possible loss of principal. The fund's investment objectives, risk, charges, and expenses must be considered carefully before investing. This and other important information is contained in the prospectus, which may be obtained at SirMayaETF.com. Please read the prospectus carefully before investing. Thanks for listening. Let's get back to today's interview. Stacey, the two biggest stocks by far in semiconductors, NVIDIA and Broadcom, among the biggest earnings growth, but yet they have been such immense laggards. Why is that the case? What was going on for a lot of this year was people were playing the quote unquote bottlenecks. So semiconductor investors love to play bottlenecks when supply gets really tight and pricing goes up. And we saw

this in memory, like these guys are guiding the EPS numbers that are higher than the stock price was a year or two ago. It's really unbelievable. And in semis, especially as earlier in the year, there was a lot of push to play the bottlenecks because I think there was a belief that there was more torque.

versus like an NVIDIA Broadcom. Although I'd say even NVIDIA Broadcom, their earnings proved to be pretty porky. But you went through this phasor, you went from playing the accelerators like the NVIDIAs and the Broadcoms a year or so back, you were playing memory, and then SETICAP, and then networking, and then optical, and then power semis, and more recently CPUs. And one at a time, these markets were sort of hitting their limits and becoming bottlenecks and the stocks were ripping. The interesting thing that I sort of observed all year is these bottleneck stocks, such as they were sort of ripping in the inventives of the Broadcom as a world that were not. And it's interesting because one of them has to be wrong. Tell me why. Why are these ripping? Because all they're going to build all this great AI stuff. Okay, wonderful. For who? Yeah. And so it felt to me like it has to normalize at some point one way or the other. It's one reason we have stocks like NVIDIA and Broadcom. Although I like...

some of the bottlenecks as well. Like I cover semi cap. I like semi cap. But I do think that this does have to normalize. We started to see a little more recently with like an NVIDIA. NVIDIA started to behave a little better recently. Broadcom not as much. And we can get into why there's like market share concerns and all kinds of things that impact these. But at some point, like I said, they're sort of creating the AI market like ultimately doesn't work. I think if they don't. So we've liked them. But you've had this very interesting dynamic between like You know, the A.I. means you got to remember investors sort of view those as we know they're going to show upside. So they're safe. So people love the shorter term money starts to use them as like funding shorts. Like you take money out of those and you buy the other by the other section. It's not like, well, I'm super negative. I want to short them. But it's like it's a stable source of funds that can be used to go buy other things that maybe are believed at the time to have higher court to the upside. So there's a lot of technical dynamics. There's other dynamics as well for Nvidia.

which may have some truth, but it's just, it's big. It's 5 trillion. It's, I don't know, eight or 9% of the S&P. And a lot of my clients, like the investors that are buying this, they've got limits in terms of how big a stock could be in the portfolio. And it's just, well, they just can't own more of it, you know? So that's, you have some of those technical issues as well that have nothing to do with the fundamentals of the stock, but it just, it is what it is. Stacy, what do you think about the memory squeeze, the broader, you know, DRAM and... NAND and then also like the hard disk drive. So I guess, yeah, descending order. Yeah, but so I don't cover those. Those are colleagues of mine. So I will avoid like stock specifics, but in general, the memory squeeze has been very interesting. People have to remember, so first of supply is very tight. Why? Demand is very short number one, but number two, people forget we're in the best memory cycle in history right now, but we are coming off the worst memory cycle since the tech bubble.

You have to remember 18 to 24 months ago, a lot of these guys had negative gross margins. It was horrible. And in that kind of environment, people would argue, well, why didn't you build more capacity? How could they? Right? You couldn't. And because memory is cyclical, they were also very hesitant to add. So they're adding now, but it takes time. They have to have a lot of confidence that the demand is going to be the before they add. There's also some other dynamics specifically on the DRAM side. And for those that don't know, there's a lot of different kinds of memory, but the two biggest pieces are what are called DRAM and NAND. DRAM is like the memory that's in your PC that runs your system stuff. And NAND is like what's in your smartphone that holds your photos, for example. They've got different characteristics and everything. But DRAM is also used in very large amounts in AI chips. And it uses a special kind of DRAM they call HBM. It stands for High Bandwidth Memory.

And what that is, it's a bunch of DRAM dies that are stacked on top of each other. There's also logic die on the bottom and you install it very close to the compute chip. It's actually on the same, they're kind of on the same piece of silicon. You install them together and you get very high bandwidth, high speeds and things like that. But the issue is because you're stacking all of these dies up, you get yield losses. Basically they're not all good. You get losses as you stack them. Not every bond you put in there is gonna be good

and some of the chips won't be good. Because of that, to make a gigabyte of high-band with memory DRAM takes three or four times as many wafers. And just to show you, chips are made. Semiconductor chips are made on slices of silicon. They look like this. It's like a 300-millimeter silicon waver. I don't know if you're watching. That looks nice. See the little rectangles on there? Those are the chips. So they make them on these things, and they cut the wafer up to make the chips. Okay. And to make a gigabyte of high-band with...

memory DRAM takes three or four of as many of those versus making a gigabyte of what's called DVR5 DRAM, like the kind of DRAM that would go into your PC or smartphone. So just because of that, if I've got a gigabyte, if I've got like a fixed amount of capacity is determined in terms of how many of these can I make in a month. Wave restarts per month is typically how they think about capacity. So if I want to make the same amount of gigabytes of HBM as DRAM, all of a sudden I've cut my capacity by two thirds to three quarters. because I need like more and more wafers to make that same gigabyte. So that right there sucks up supply on top of the then. So that is something that is new. And in memory, you know, in anything like, this time is different is always a dangerous like kind of statement. That is different this time. This dynamic of high end with memory taking up more wafer capacity than standard DRAM is new. So that's been a driver of the tight supply as well on top of the very strong demand. On the NAND side, For a while it was just the NAND guys decided they like to earn money for a change, and so they didn't want to add any capacity. They are adding capacity now, but it turns out you actually, as you build out this AI infrastructure, it drags a lot of NAND storage basically along with it. You use it for things that I don't want to go into, the things that are called KVCache, and these models get really, really big, and they've got a lot of what are called weights, a lot of parameters. You have to store those weights.

especially generative AI that is generating, you know, it's out of its videos or whatever content that content has to be stored somewhere. So now there's also a lot of demand as well as very tight supply. And so they're out in capacity now, but it takes years. If you're building new factories, it takes years. So it doesn't alleviate any time soon. And you got to have the hard

drives. They were getting a higher multiple for a long time because they were viewed as being, it was a duopoly basically. It was like, that the two primary players are Western Digital and Seagate, but they were getting a higher multiple than the memory guys mostly because of that duopoly. And now I think it was Toshiba the other day talked about they want to start adding capacities that hit stocks, although Toshiba is not that big. And, you know, but I mean storage demand is very strong. It's even for hard drives. I read somewhere the other day that even like maybe magnetic tape might be making a comeback now, which tells you something. If that's true. So just in general, I would say just AI is dry. It drives a ton of demand for memory and storage.

And by the way, if you think, if you look inside one of these racks, so like NVIDIA sells a big rack, it's called like an NVL72 and it's got your 72 GPUs and 36 CPUs, a bunch of other stuff in there. But if you look at the semiconductor content that's in it, not from a dollar standpoint, but from a wafer area standpoint, memory like DRAM and then it's like 80% of the silicon wafer area inside that rack is memory. So as they sell the stuff, you just need a ton of memories. The supply is not there. And so they got to add. And that takes a lot of time. And we'll see what happens as it gets added. It'll be twenty seven twenty eight before you start to see material amounts of capacity start to come on. And then we'll see what happens with pricing. So the the semiconductor fabs that make logic chips, memory chips as well, they buy from semi cap equipment companies. There are five of them. You cover three of them.

There's more than five big ones. Five big ones. Five big ones. Five big ones. Five big ones. You cover three of them. What is your outlook? Which one is your favorite and why? Yeah, so we like semi-cap. And my background, by the way, is in semi-cap. Way back in a prior life, I used to build semiconductor manufacturing equipment. That's where I came from. So I've always had a soft spot for the space. My PhD is in plasma etch, actually. Wow. From long ago. But I really do like this basis. Again, we need more. supply and it's the same kind of thing like the memory guys, for example, are building capacities. They have to buy tools, right? But we've sort of done the work, like you can sort of work backwards from, you know, what is your data center for? Because how many gigawatts per year are going to

get built, right? And you can translate that back to, well, okay, so how many racks am I going to need? And then you can translate that, well, how much silicon is in a rack? And you can translate that back to, well, how much capacity do I have to add? And so therefore, how much do I need to spend on WFE?

WFE, by the way, stands for Wafer Fabrication Equipment. People use that as sort of a representation of the whole industry, of a whole sunny cabin today. How much is the energy spending on tools? What's your WFE forecast? We will probably wind up spending 150 billion plus on WFE equipment this year in 2026. We are over 200 billion next year and I think over 250 in 2028. And if you do the math on, you look out to the end of the decade and look at the forecast and say, how much data center capacity do people generally think is going to be online? I have no problem thinking that W, if he needs to go 300 billion or well over. I actually suspect that if there were no constraints, we would be there very quickly. The issue is right now, clean rooms. So a semiconductor factory is called a FAB and it consists of what's called the shell. That's the building.

Inside the building are clean rooms and the equipment goes in the clean room. And it's called a clean room because it has to be very, very clean, right? You can't get particles and dirt. These are cleaner than like hospital rooms, right? And right now, there are no clean rooms. There's a clean room shortage. So before they sell the tools, they have to build the clean room. So back to my earlier point, there were supply kinks drinks that's stopping things from completely getting nuts, right? I suspect that there is demand, like we would be shipping over 300 billion like next year if clean rooms were like unlimited. And if the semi-cap companies themselves had enough capacity though, they don't, right? So I don't know if we can get like upside too much up to this, say 250 and 28, we'll see if we can get there. But I have no problem given how much data center capacity, at least assuming the data centers get built.

No problem signing up for a 300 billion plus like WT into the end of the decade. And I suspect if clean rooms were unconstrained and capacity was unconstrained, we would be there earlier. But because of that, I'm actually quite bullish on the semi-cap themes. I like semi-cap in terms of which one

do we love? I mean, the correlations are very high. So if you want to own them, I would say you could own the group, right? You don't give up that much. They have different exposures say to end markets. So one guy might have more logic, one guy might have more DRAM that sort of thing. You know, but I cover AMAT, LAM, and KLIC. I like, we like them all. Like if I had to lean a little toward them, maybe it's AMAT a little bit just cause they have more DRAM exposure. We need a lot of DRAM to make the HBM on it. And it's a little cheaper, but I think you get on any of them or all of them if you want it. I've got colleagues of mine that cover.

Some of the other stocks of the day cover Tokyo Electron and ASMEL. The big five are ASMEL, Tokyo Electron, Applied Materials, LAM Research, and KLA. So my colleagues cover the other ones. They also cover some of the other smaller ones in the space, Bassie and some of the Chinese names and some of the others. So I thought that LAM Research had the biggest exposure to memory, but then you tell me about applied materials as the highest exposure to DRAM specifically. So is it that AMETS overweight DRAM and underweight NAND? Yes, that's exactly it. LAM is overweight, NAND. Although LAM is really interesting, by the way, because if you go back five years, 10 years, there was a point in time where LAM was probably 60 to 80% memory. And of that 60, 80%, like two thirds to three quarters of it was NAND. So LAM was always used as like the NAND flash semi-cap company. And they still have, I think 25 or 30 cents out of every dollar spent on NAND equipment.

goes to land. But they've done a really good job over the years of diversify. So as NAND is still important and they still have the biggest percentage coming from NAND, they've done a really good job building up like their foundry and their logic and even some VRAM, right? So they've done a really good job of diversifying and it's, I'll give you an example, like during COVID, we had a bit of a dip, especially in NAND flash. NAND went through a desert during the peak of COVID, the industry was spending 20 billion plus a year on NAND equipment. and then it went to like five, like it collapsed. And Lamb had this one quarter where they guided EPS for the following quarter to like 5. They did a 10 for one split since then. So today that would be 50. But they guided five bucks for this one quarter coming up and they actually wound up beating the number. And so it was a 20

annualized EPS guide for a purportedly NAND focused semi-cap company with effectively no NAND in the guidance. It was close to zero.

And it was not great of them, but it was fine versus like if you went back to prior memory down cycles in prior years, you know, go back five years or 10 years, they would have been absolutely bleeding at the bottom of the cycle. And in this case, like it was fine, right? They were doing just fine. So they've done a really good job over the years of diversifying away, even though man is still very important for them. Stacey, I'd say that generalist investors who are interested in semiconductors and I include myself in that category, I think we look at DRAM and NAND as kind of the same thing because that's memory and there's been a giant memory squeeze because of AI. Tell us about the differences, not just technically, but in the ecosystem. And if there is a split and the divergence between the two, how are we going to see that split? So generalists should be aware of this split right now, the difference. First of all, you got to remember.

A lot of the memory companies make both. So Samsung and Heineck's and Micron are like the big man and they made both, but they're not the same thing. So from a technical standpoint, DRAM costs more per bit. In memory, basically with the cost is related to how big is the cell, because in memory, you just want to squeeze as many bits per square millimeter of silicon as you possibly can. So the DRAM cells are bigger and the structure is totally different. The technology is different, but per bit DRAM costs more. DRM is also what's called volatile. So when it's in your PC or whatever and it's running now, it's volatile. Your PC is writing to that DRM at every 100 milliseconds. It has to rewrite it. And when you pull the plug out of the wall, everything that was in the memory on DRM goes away. Man flash is a little different. It's called, it's non-volatile. It works by storing charges.

in a cell to register a bit one or zero, is there chargers or not? It gets more complicated than that, I'm simplifying. But it's non-volatile, so once you read to it, once you write to it, it stays there. They pulled NAND chips out of fires and been able to get the date off. It's actually a remarkable technology. Per bit NAND is much cheaper, because the density is much where you get a lot more bits per square millimeter. It is however much

slower to read to write, especially to write to it than you, which is why they don't use it in your PC and they use it instead. It's much slower. But my man is used to like, say, store your photos or something in here, right? But this phone also has DRAM in it to run the software, to help run the software. But those are the differences in terms of technology. From an industry structure standpoint, there are more players in NAND. And so the competitive dynamics in NAND sometimes are a little worse. And we've seen that in terms of some of the margin profiles in some of the names. And you've got the big three, but you've also got...

There's there's solid dime. And there is kiaxia. And you've got some of the Chinese guys. Y and T.C. makes manned. When D run, you've got the big three. And then you've got CXMT, which is a Chinese player. So there's like four guys sort of in D run. You got probably six or seven or eight men. So more complication than that. Stacey, tell me about your industry estimates that you have for WFE and how that. plays into the estimates for like a LAM research or an AMAT. Are the numbers that you have for LAM and AMAT kind of a conservative hedge relative to that outlook on WFE? Oh, I mean, they are consistent with our published forecasts on WFE, which I can't remember the exact numbers. It's like, I can't 154 billion or something for this year. And we're at 204 for next year. 204.

and then something like 256 or 258, a little over 250 for 2028. That's our modeling. Our published numbers for our covered companies are consistent with those WP forecasts. We've done the work, like I said, on if the data is going to be able to happen, beyond that, well, we need more. I suspect we will need higher WP into the end of the decade. I suspect demand is high enough to do it earlier. I just don't know if the clean rooms are available to enable it. Because before I can ship a tool, I need to have somewhere to put it. Yes. And you need to build those, the clean rooms are full. So they've got all built in. If you listen to Micron's, Micron just reported earnings, they actually talk about CapEx is going up and they're going to be spending more on the facilities because they have to build clean rooms. So that'll be growing. I think it's all growing for them, but.

facility spend is going to grow faster, I think than the equipment for their next fiscal year, because they have to build the clean rooms. Once they

build the clean rooms, then presumably you'd fill the clean rooms up with equipment. I want to ask you about the threats to Nvidia, whether they're the, you know, the new IPOs, the startup companies, Broadcom having custom ASICs or AMD, which would you say poses the most serious threat to Nvidia's dominance in GPUs? To be honest, I would call them perceived threats. Right. And so you mentioned one broad one, which is competition. This is semiconductors. Like there's always somebody looking to eat your lunch. Right. So it is imperative on you as a leader to make sure you stay there. And how do you do that? You invest and you leverage other aspects of whatever you can do, create your mode. And so NVIDIA is investing like crazy. And they have software that people call a CUDA. It's not just the program environment. It's also the broad ecosystem that builds around their hardware. It's.

a variety of different types of silicon. It's not just the GPU, it's GPUs and it's LPUs and it's CPUs and it's networking and all this stuff. And then look, even their balance sheet is a moat. They're investing broadly in the ecosystem. They're taking equity stakes in players. You know, people use the word backstopping. They're not necessarily writing a bunch of checks, but they're sort of like lending out their credit rating, right? To ensure that the infrastructure gets built in terms of that. That's a moat. I'd say the one that people probably look at most is competition. Because there's lots of folks you've got guys like broad you've got guys within in GPUs like AMD you have Asic play ASIC by the way means custom chip and stands for application specific integrated circuit when you're ASIC ASIC Custom chip. So you got players like Broadcom or even like MediaTek and Marvell and others that are making custom chips. There's lots of them Of course they are this is a you know, we're getting up to a trillion dollar market just remain of course there's competition, right?

My general view on that, by the way, at least at this point has been it's the wrong question to ask. And I understand why people ask it. Seven investors love to play the competition narrative and it does move the stocks. But from a fundamental standpoint, is it big still or is it not? If it's big, I think everybody's fine. And if it's not, I think everybody's screwed. Right now it's big and you can look at the numbers for next year. Nvidia said that they're basically going to do 700 billion dollars. 70% plus growth. We get you like

close to 700 billion. And they could do even more if they get more upside, right? Broadcom said they'll do 115, 115 in AI revenue next year. And again, they could do more. AMD, if they can ramp the rats, I mean, they could do, you know, 40, 50 billion dollars, right? In this, I don't know where the consensus is for MediaTek. I mean, 15 billion, maybe more for MediaTek. Marvell, we don't cover, but it could be 10 billion. You got even guys like Qualcomm could do five next year.

Everybody's doing fine. I'm not even convinced. I think if you were to calculate the percentages, like the market you're presented on those numbers, I don't even think they're moving very much. So I actually fundamentally, I don't really worry about market share for any of my guys. Like NVIDIA is big enough where they can handle it and the other guys like at A&B or even a Qualcomm are small enough where if they just take a little bit, like it's meaningful for them, but it's not that much relative to the whole. And is the opportunity bigger or is it not? I think it's big. Tell me about Intel and your work there, your recent conversation. Yeah. So Intel is interesting. So I like to say, and I've written this, I feel better about Intel right now than I have maybe ever, which is a very, very low bar because I've literally made my career being negative on it. That was literally the call that made my career. It was a good call for a very long time. 15 years, it was the gift that keeps on giving. It really was. That being said, I don't hate it right now, which is new for me.

It really is. That's saying a lot. They've got a few things going for the right. They've gotten lucky. The server market is saving their behind right now. Their server products by their own admission are not really competitive. But right now it doesn't matter. Demand is so strong the customers are buying anything they can get their hands on. Intel themselves had a couple of quarters ago. They had 200 basis points of gross margin upside because they had server products that they rode off. They rode them off to zero. They thought they were going to have to scrap them because they were bad. Customers said, we don't care, we'll take them. That's how strong demand is. So look, take lucky over good. Like it's God knows they were due for a break. So that's good. The other is I think the narrative is going their way. Like supply is tight and you know, they have this foundry business they're trying to, I still think it's not gonna be for quite a few years before you see

any sort of material foundry revenue, but our customers probably giving them a harder look. Yeah, sure. You want other sources? They'll get packaging revenue. We didn't talk about what packaging is, but that's something else that has to be, It's lower margin, but they'll get packaging revenue. They've got some decent IP there. Trump wants the stock to go up, that doesn't hurt, right? And the CEO of Lippu Tan, I actually like a lot. I think Pat Gelsinger, who you replaced. Pat, there was nothing wrong with Pat's strategy, but I think Pat's execution and communication to that strategy was horrendous. Like he came in and it was like everything's perfect and everything's fixed. He got his ass handed to him.

Right? I mean, I don't know who comes into a turnaround and acts like Pollyanna. So at least Leapoo is like realistic. You know, he's getting the cost structure in the place. Pat hired 21,000 people and he had to fire them all. Right? So Leapoo got the cost structure in place. And you know, Leapoo also, he's a legend in the industry. He knows everyone in the industry. He could pick up his phone and call anybody. He knows how to run a foundry. He was on smicks board for 20 years. And he knows what the ecosystem needs. He's got a bunch of startups that are all like fabulous like customers of TSMC. So he knows what the customers want and he knows everybody. So that's all those are all good things. I still think it's going to be a slog for them. And I think they know that too, you know, but, but look, I think things are better for them than they have been in quite a while. And I'll leave it at that. I don't hate it, which is for me is saying something. You have a flash card that.

Even though I understand you're very positive on the semiconductor sector for good reason, I think that there are some things that traditionally through your experience in the semiconductor industry have found that they are a little bit of a, not a red flag, but a red light. So the semiconductor inventory is going up, channel inventory is going up. How many of these do you think are sending you a bear signal versus oh. Historically, this is bare signal, but like this time is different kind of yes, you're referring we do a cycle track for like every quarter. Yeah, it's got like traffic lights red yellow and green for we track like growth expectations versus normal season L our expectations higher low all speak I rather see them low versus high channel inventories We look at like the distributors and and our inventories higher

or low inventories on on semiconductor company balance sheets We track valuations, we track earnings revisions, all that kind of stuff. So the ones you're probably pointing, it's probably inventories is the biggie. And they're very high in the channel. However, they've been high for like four or five years now. And this is one problem I will admit I'm having a lot of the traditional cyclical of signposts you would see. We haven't really had a typical cycle quote of what for almost 10 years.

What do you mean by that? I'll give you an example. In May of 2018, now it's 2026, it's almost 10 years, like eight years ago, right? I remember I downgraded Texas insurance and analog devices in May of 2018 because I thought numbers were too high and typical stuff, right? And I was right in the sense that like by the end of 2018, my numbers had fallen by like 40%. But the stocks wound up higher than they were the day I downgraded them. And both of us went about... The reason is in 2018, what was everybody playing for into the end of 2018 to 2019? This is trade war 1.0. Everybody was playing for a trade deal. So everybody would cut numbers and people would say, oh, well, that's the last cut. Now we're going to get the trade deal. And so the stocks are, because you have to have recevies or anticipatory. So that was for most of 2019, even as numbers were getting cut, the trade deal was keeping on going. And then what happened? COVID. So we had everything collapse at the shutdowns, but then it turns out, We were all like, I get our houses, anything work from home, study from home, play from home, just like demand ripped. And there was a bunch of supply shortage because everybody had shut down their factories and it was like, oh, shit, like now what do we do? We got to start them up again. So that caused a lot of problem. And so everything ripped on the back of COVID, right? We had the COVID overhang, but even during the COVID overhang, we had a lot of these names, like numbers got cut a bunch.

But in many cases, the stocks didn't really go down that much. Like the denominator got smaller and the multiple went up and the stocks didn't really move that much, but ordinarily they would have a lot, but they didn't. And now we have, it's Trump 2.0 and we had tariffs that crushed everything and then the tariffs went away and everything ripped again. And now we've got AI, which is, I've never seen a demand cycle, driven cycle of this

strength before. We haven't had a normal cycle. I don't know how representative these historically typical cyclical indicators are anymore. According to the old cycles, yes, really high inventories, like that would be a red flag, but they've been high for four or five years, like this is the new normal, like I don't know. Stacy, what was things you saw in 2000 that were classic signs of a top? And are you seeing anything now? 2000 was a little before my time. I started this job in April of 2008, three weeks after Bear Stearns failed. So the financial crisis was where I was tested in fire. I was in grad school in 2000. However, In 2000, I would say there's a few things. One is we talked about stockpiling and over there, you know, dark GPUs. I don't think there's a lot. There was dark fiber. We've got dark fibers sitting in the ground today for demand that was built up for demand that never showed up. So that was one thing that was different. The other is the valuations were much higher. I'm going to make, I don't remember what it was, but Cisco was 200 times forward earnings, whatever the hell it was in 2000, right? Again, and then he is probably at 12 times, right? So.

Like we don't have the same kind of multiple driven, like if you're worried about a bubble, it's definitely on the multiple side. You have to argue why our earnings in a bubble. And even there, like the multiples are not that high. So that's one big difference, I think between the two. So one is the build out ahead of demand. I think actually right now demand is, nobody has enough. Like so we're actually building out with demand versus building out ahead of demand. And then the valuations are much different, I think. And that's a lot of city. Like I said, I didn't even know what a stock was in the year 2000. That wasn't on my mind.

Back then well, so Stacy to sum up Could you say you just? What is your perspective on on semi stocks right now? They have run up the past month of course from there, you know the big a little panic we had in in June and I believe there are two stocks They're kind of your favorite, right? I get the nervousness. I understand but I still think fundamentals overall are getting better. Not worse Again, the sector is still up a bunch here today. If you've been there all along I mean clearly if we got here monotonically everybody would feel fine Never happens like you do this, but I'm still fairly constructive overall You know, what do we like? I mean we like the AI the traditional AI names. I also like semi cap on the CPU side We've liked

AMD so I mean those are your Nvidia Broadcom semi cap AMD is kind of where I've been spending most of my time at least. Stacey, thank you so much. People would find you on LinkedIn and you're also on Twitter at sRaskin. Really appreciate it. Thank you everyone for watching. Please leave a rating and review for Monetary Matters on Apple Podcasts and Spotify and don't forget to subscribe to the Monetary Matters YouTube channel.

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